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Propagation velocity of landslide-induced liquefaction and entrainment of overridden loose, saturated sediments

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ABSTRACT

Loose saturated granular materials are particularly susceptible to instability, resulting in deviatoric strain softening, and static liquefaction. When instability occurs in the context of a landslide, the consequences in terms of the mobility of the debris and risk to life and property can be catastrophic. Physical model landslides initiated in a geotechnical centrifuge under rising groundwater conditions were used to trigger instability and static liquefaction. Four experiments with a loose contractile soil and two experiments with a dense dilative soil were performed. The velocity of propagation of the liquefaction front within the loose granular soils at the base of a landslide was quantified using a dense sensor network of pore water pressure sensors and high-speed imaging. On triggering of a landslide, a localized toe failure was observed to shear and liquefy the soil at the base of the landslide. However, the velocity of this initial failure (0.5 m/s) was an order of magnitude slower than the subsequent 4.2 m/s propagation velocity of the liquefaction front. These experiments demonstrated and quantified how a localized failure onto a liquefiable deposit may propagate liquefaction much farther than simply the runout of the localized failure, and highlight the potential implications and consequences of such an occurrence.

1. Introduction

Landslide runout analyses that are carefully calibrated through simulations of the motion of past landslides can be used to predict the motion of potential future landslides, informing landside risk assessment and design of mitigation strategies (e.g. McDougall, 2017). Instability, defined as a behaviour in which large plastic strains are generated rapidly due to the inability of a soil element to sustain a given load or stress at a state of stress on, or in certain circumstances, before reaching the failure envelope (Chu et al., 2015), can have a significant impact on landslide mobility. Loose saturated granular materials are particularly susceptible to instability, resulting in significant deviatoric strain softening, and in extreme cases, static liquefaction (e.g. Lade, 1992; Chu et al., 2003; Yamamuro and Lade, 1997; Wanatowski and Chu, 2007; Baki et al., 2012). When instability occurs in the context of a landslide, the consequences in terms of the mobility of the debris and risk to life and property can be catastrophic.

In this paper, we explore the scenario of a landslide impacting, liquefying, and entraining loose saturated sediments. This scenario could represent the entry of a landslide into loose saturated liquefiable tailings in a canyon-fill tailings storage facility, a landslide overriding and entraining loose saturated valley floor sediments, or an internal berm failure into dredged and hydraulically placed loose saturated sediments.

Some possibilities for the behaviour of a landslide impacting and overriding a sediment layer are illustrated in Fig. 1. Fig. 1a illustrates the case of a landslide triggered at the toe of a slope with the debris flowing over a dilative (i.e. non-liquefiable) and low erodibility soil bed. If this layer resists entrainment, the distal reach of the landslide, x_{max}, and time history of instantaneous slide velocity, v_s, will be simply a function of the rheology of the deformed landslide debris. This scenario is the most commonly considered case in physical or numerical modelling of landslide runout and is well suited to the use of simulations of the motion of past landslides to predict the motion of potential future landslides.

A more complex scenario, discussed by Sassa (1985) in the context of debris flows, is illustrated in Fig. 1b in which the soil at the base of the slope immediately below the sliding mass liquefies and becomes entrained in the landslide debris. In this scenario, entrainment can be the result of the conventional mechanism of basal scouring seen commonly in scenarios of overriding by frictional or collisional flows (e.

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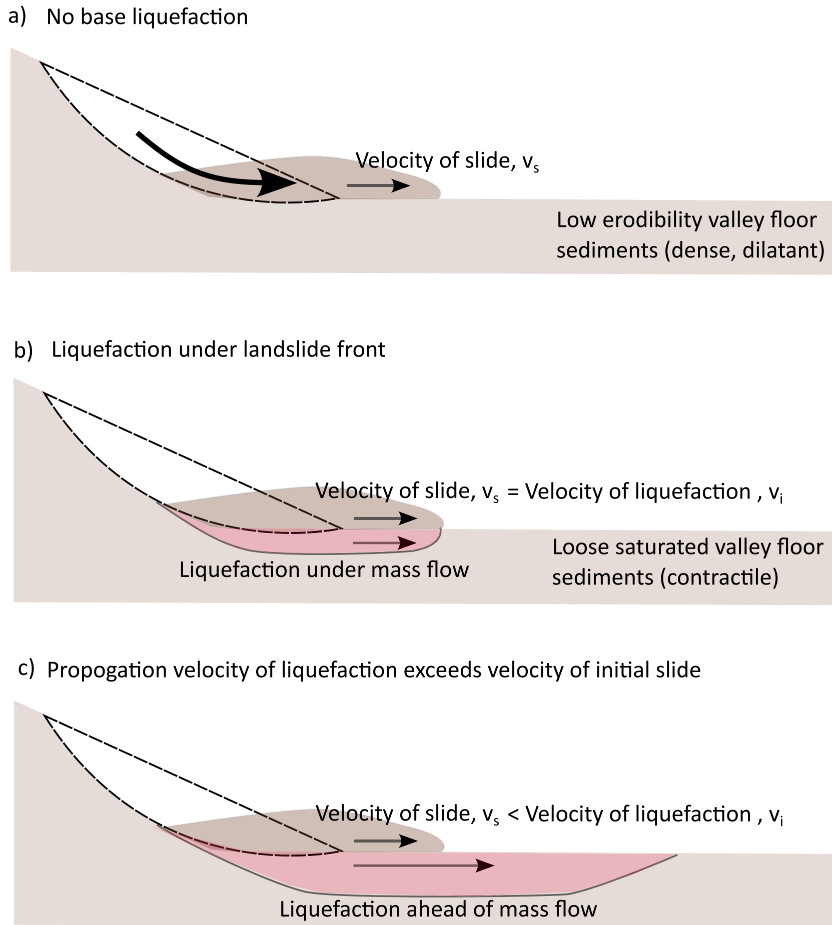

Fig. 1. Scenarios illustrating the influence of liquefaction of valley floor sediments on the mobility of landslide debris. a) If no liquefaction occurs, the mobility of the debris will be strictly a function of the rheology of the debris; b) case in which liquefacion occurs only under the front of the landslide ($v_s = v_i$), and c) hypothetical case in which the propagation velocity of instability exceeds the initial velocity of the slide triggering liquefaction ($v_i > v_s$).

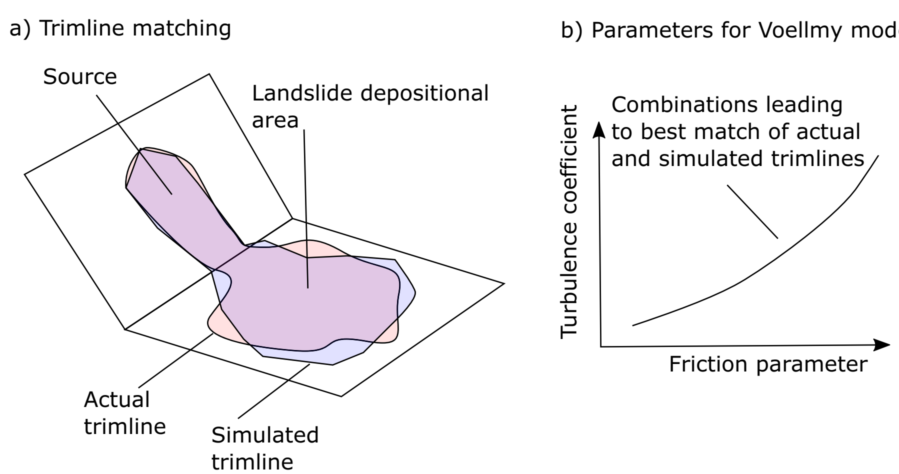

Fig. 2. Calibration of landslide runout models based on trimline matching of motion of past events; a) actual and simulated trimlines can be used to quantify the degree of match for each set of parameters used in a simulation, leading to b) ranges of parameters providing the best match being identified.

g. Song and Choi, 2021; Choi and Song, 2023; Liu et al., 2023; Zheng et al., 2023), or by the less well documented mechanism of fontal plowing (e.g. Shen et al., 2019; Choi and Nikooei, 2023). Field evidence of plowing and impact liquefaction have been discovered in loess landslides within the terrace sediments of the Jing River on the South Jingyang Platform, Jingyang, Xi'an, China by Peng et al. (2018), providing context for the high mobility of landslides observed in the area. Recent attempts to reproduce the mechanism of impact liquefaction in the laboratory by Hu et al. (2023) illustrate that dry debris can fluidize and entrain bed material by plowing. However, while pore pressure sensors embedded in their physical model tests recorded positive excess pore water pressures during shearing, they were not of sufficient magnitude to liquefy the bed material for the conditions tested. These observations illustrate the complexity of the mechanism of impact

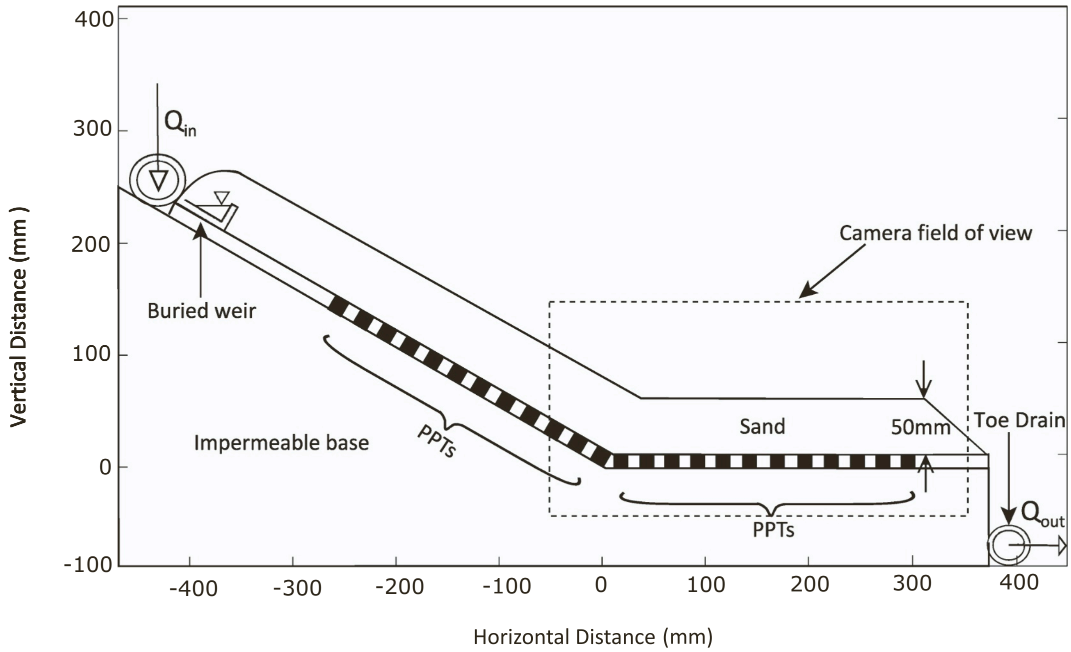

Fig. 3. Centrifuge model in which a thin soil layer will be brought to failure under raising groundwater conditions. A groundwater flux of Q_{in} is provided at the top of the 30° sloping portion of the model, and a horizontal region of valley floor sediments if provided to serve as either a dilatant or liquefiable layer. A dense network of pore pressure transducers (PPT) are included to infer propagation velocity of instability and inform visual observation of slope behaviour captured using a high speed camera.

Table 1
Testing summary.

Test number	Initial void ratio	Thickness at 1 g (mm)	Thickness at 20 g (mm)	Flow Rate at Failure (mL/min)	Model Outcome
Loose 1	1.45	52.6	37.3	382.4	Liquefaction
Loose 2	1.62	56.7	37.7	259.4	Liquefaction
Loose 3	1.17	50.3	49.7	416.3	Liquefaction
Loose 4	1.79	58.8	36	222.7	Liquefaction
Dense 1	0.7	49.9	49.9	450.2	Erosion of toe
Dense 2	0.79	49.9	49.9	517.9	Erosion of toe

liquefaction, and highlight that additional research is needed to explore the conditions required for liquefaction to occur.

Recognizing that static liquefaction landslides were difficult to achieve in physical model tests (Take and Beddoe, 2014), Take and Beddoe (2014) illustrated that the presence of a loose layer of soil at the base of the landslide (i.e. valley floor sediments below the toe of the slope) introduces a soil region particularly prone to static liquefaction as this scenario fulfills the criteria required for pre-failure instability: a) loose contractive soil, co-located in an area that is b) fully saturated, and c) within reach of a shear trigger (i.e. being overridden by the landslide). In this scenario, the landslide overrides the loose contractile soil and, if it is assumed that entrainment only occurs immediately under the front of the landslide, the propagation velocity of the instability (v_i) occurs at, and is limited by, the velocity of the overriding landslide, v_s. While a convenient assumption for the inclusion of entrainment processes into dynamic landslide analyses, it is hypothesized in this paper that the propagation velocity of instability could, in certain circumstances, exceed the velocity of the initial shear impulse (i.e. toe failure).

The shear wave velocity in soils at small strains, measured using geophysics Instrumentation, is on the order of hundreds of meters per second. In contrast, liquefaction is a large strain phenomenal (often taken as 3.75% strain in element tests). It is therefore likely that the propagation velocity of the large amplitude shear wave imposed by the shear impulse from the overriding landslide will be limited to a velocity significantly slower than the shear wave velocity. However, if it is possible for the propagation velocity of the instability, v_i, to exceed the velocity of the landslide, the rate of growth of the size of the landslide could exceed the velocity of any single point in the sliding mass (Fig. 1c). The maximum distal reach of the landslide in this scenario would not simply be a function of the rheology of the deformed landslide debris, but also the velocity of instability, v_i. Such a scenario, if possible, would have significant consequences on the rheological parameters derived from a back-analysis of past landslides influencing predictions of future events.

The typical workflow for parameter calibration of landslide runout models is illustrated in Fig. 2 inspired by the comprehensive review of the current state of the art and challenges facing landslide runout prediction conducted by McDougall (2017). Using the commonly adopted Voellmy rheology as an example, the shear stress acting on the base of an element of the landslide can be described as a function of two parameters–- a friction coefficient and a turbulence parameter. Combinations of these two parameters can be used in a semi-empirical fashion to attempt to match the trim line of the model to the observed landslide debris (Fig. 2b) and identify parameter combinations that result in the best match (Fig. 2a). These parameter combinations could be refined further if data or field evidence of landslide velocity at a point are available. However, if it is theoretically possible for the velocity of instability in the base soil to exceed the initial landslide velocity, parameters derived from such a back analysis will not be representative of the actual rheology of the soil.

While the field evidence of Peng et al. (2018) indicates that impact liquefaction can occur, the lack of field data capturing the details of the mechanism through which it occurs leads to uncertainties regarding the exact process involved. Specifically, in the absence of any data observing the propagation velocity of instability in static liquefaction landslides, the possibility that v_i may differ from v_s remains an untested hypothesis. Given the significant implications of this hypothesis for runout analyses, the objective of this paper is to conduct physical model testing in a geotechnical centrifuge in which landslides experiencing static liquefaction are triggered from an initial toe failure to investigate the relative

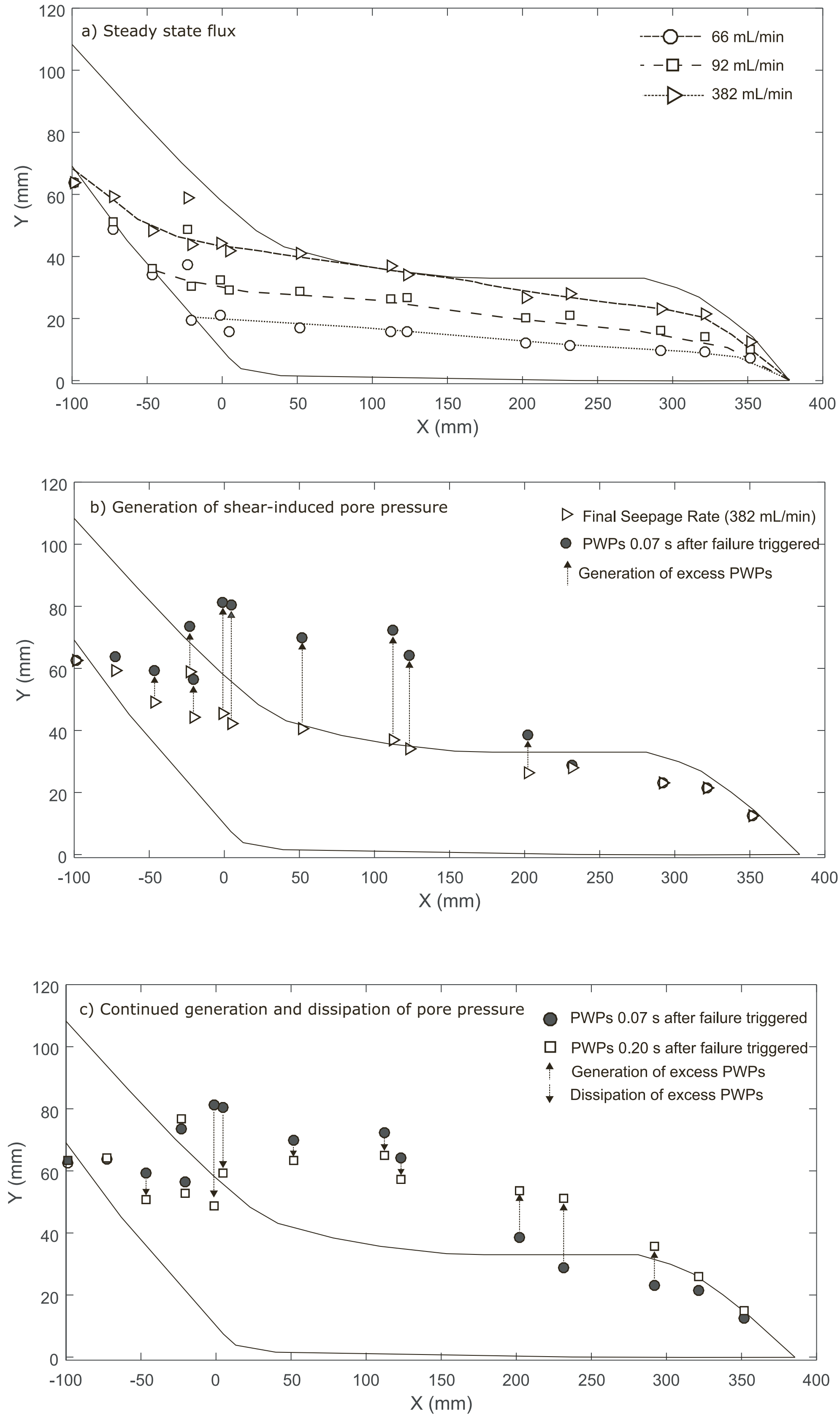

Fig. 4. Pore pressure response of soil in test Loose 1 (L1), during a) step-wise increase in steady state seepage flux, slowly generating a saturated zone of liquefiable soil in the valley floor sediments, b) transient pore water pressures during the first 0.07 s of liquefaction of valley sediments, and c) continued generation and dissipation of shear-induced pore pressures during the subsequent 0.13 s.

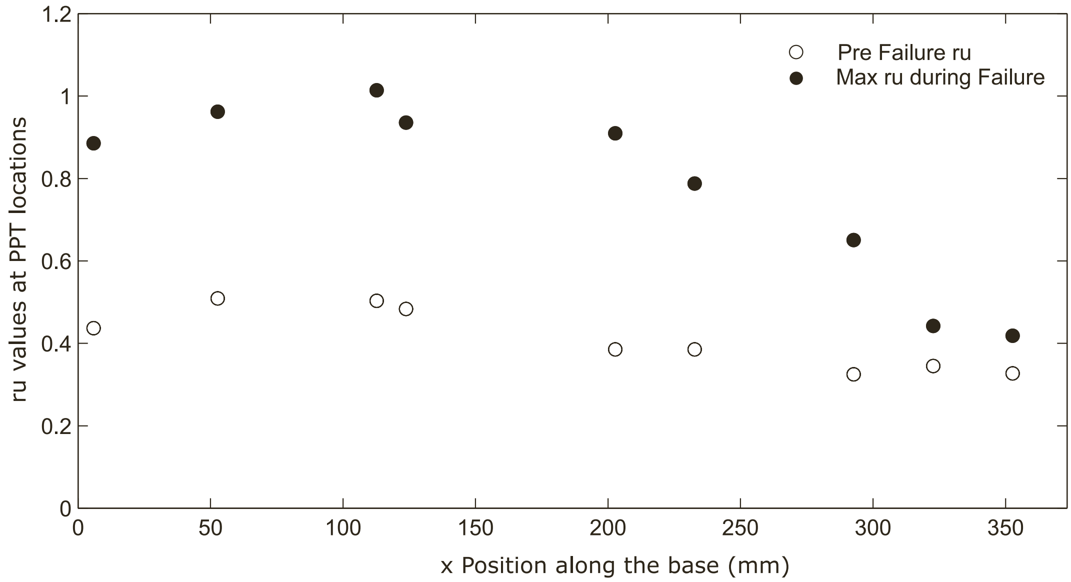

Fig. 5. Assessment of magnitude of excess pore water pressures immediately before and aer instability expressed as the pore pressure ratio, Ru, against measurement location along the horizontal base of the model. Values at or approaching unity during instability confirm the liquefaction of the valley floor sediments during shearing.

velocity of the initial failure event and the propagation of instability in a loose, saturated basal soil layer. In the remainder of this paper, we describe the physical model, the boundary conditions imposed and the instrumentation used to measure and compare the velocity of the landslide to the velocity of instability as captured in measurements of excess pore pressure captured along a densely spaced linear network of pore pressure sensors. These two data sources will then be used to explore whether it may be possible for the velocity of instability to exceed the velocity of an overriding landslide.

2. Physical model

2.1. Geometry and boundary conditions

Reduced-scale physical models of slope stability processes do not represent scaled field behaviour when capillary stresses dominate over self-weight stresses (i.e. the sand castle effect leading to the possibility of unsupported vertical slopes of unsaturated soil). The technique of geotechnical centrifuge modelling in which the self-weight body stress of particles within a small scale model is enhanced with depth offers a unique mechanism to overcome this limitation and regain the appropriate balance between these two stresses. The ability to precisely control boundary conditions has led to centrifuge models of landslide processes being used to better understand the mechanisms that contribute to mass wasting events and the resultant mobility of the landslide debris (e.g. Take et al., 2004; Ng, 2008; Askarinejad et al., 2015; Take and Beddoe, 2014; Beddoe and Take, 2016). Of particular relevance to the present study, the work of Take and Beddoe (2014) illustrated that the presence of a loose saturated region in the valley floor sediments at the base of a slope increases the susceptibility to a slide to flow mechanism of liquefaction. Further, Beddoe and Take (2015) illustrated that the runout distance of a landslide on a 20° slope may in certain circumstances exceed that of a 30° slope as the former requires less groundwater flow to trigger liquefaction. The effect of antecedent rainfall in terms of antecedent groundwater conditions prior to triggering of instability under rainfall infiltration was investigated by Take et al. (2015). Their results indicated that groundwater levels just below that required to cause failure, led to landslide runout distances three times further than in a nominally identical slope brought to failure under rainfall alone.

The present study builds on this body of work by adopting a similar geometry to these past studies, as shown in Fig. 3. The model geometry represents a thin (50 mm) soil layer inclined at 30° to the horizontal overlaying an impermeable bedrock contact. When tested at 20 g, the model corresponds to a soil thickness of 1.0 m at prototype scale. The bedrock consisted of an aluminum plate treated with a waterproof anti-slip frictional coating and a moldable sealing putty (Duxseal) was used to ensure a no-flow boundary between the viewing window and the aluminum plates. Following the work of Take and Beddoe (2014), the geometry of the model includes a horizontal layer of soil at the toe of the slope to provide a basal layer of loose saturated sand susceptible to static liquefaction. The change in hydraulic gradient from the inclined portion of the model to the base of the slope ensures that the height of groundwater flow in potentially liquefiable basal soil layer will be at or near the top of the layer at the time of landslide triggering ensuring the required high degree of saturation for the liquefiable layer. Triggering of instability in the model was then conducted through introducing a groundwater flux at the top of the slope, denoted as Q_{in}, in Fig. 3 using a variable metered pump to precisely control the flow rate. Water from the pumps flowed through a perforated pipe to the crest of the slope, where it infiltrated the soil and contacted a buried weir precision machined so that water would be distributed evenly along the width of the soil in the model accounting for the slight curve in the acceleration field of the centrifuge. The water that made its way through the soil and drained out the toe (Q_{out}) was released to the centrifuge chamber. Groundwater flow through the model was increased incrementally resulting in a controlled increase in the groundwater table. During each increment, the flow rates were increased once steady state conditions had been achieved. Steady state conditions were determined based upon stabilization of pore pressure readings of pressure sensors imbedded along the base of the model at the sand-bedrock interface (full details on the instrumentation scheme will be presented later). Flow through the model was continued until a landslide occurred or the flow rate of the pumps were exceeded.

In centrifuge testing, the scaling relationship for time is recognized to be dependent on the phenomena under investigation. For diffusion processes such as consolidation, time is considered to scale as $1/N^2$ (where N is the g-level) while for inertial or dynamic behaviour, it is traditionally interpreted to scale as 1/N (e.g. Dewoolkar et al., 1999; Taylor, 1995; Madabhushi, 2014). The conflict between the time scales of dynamic and diffusion events is typically overcome in centrifuge

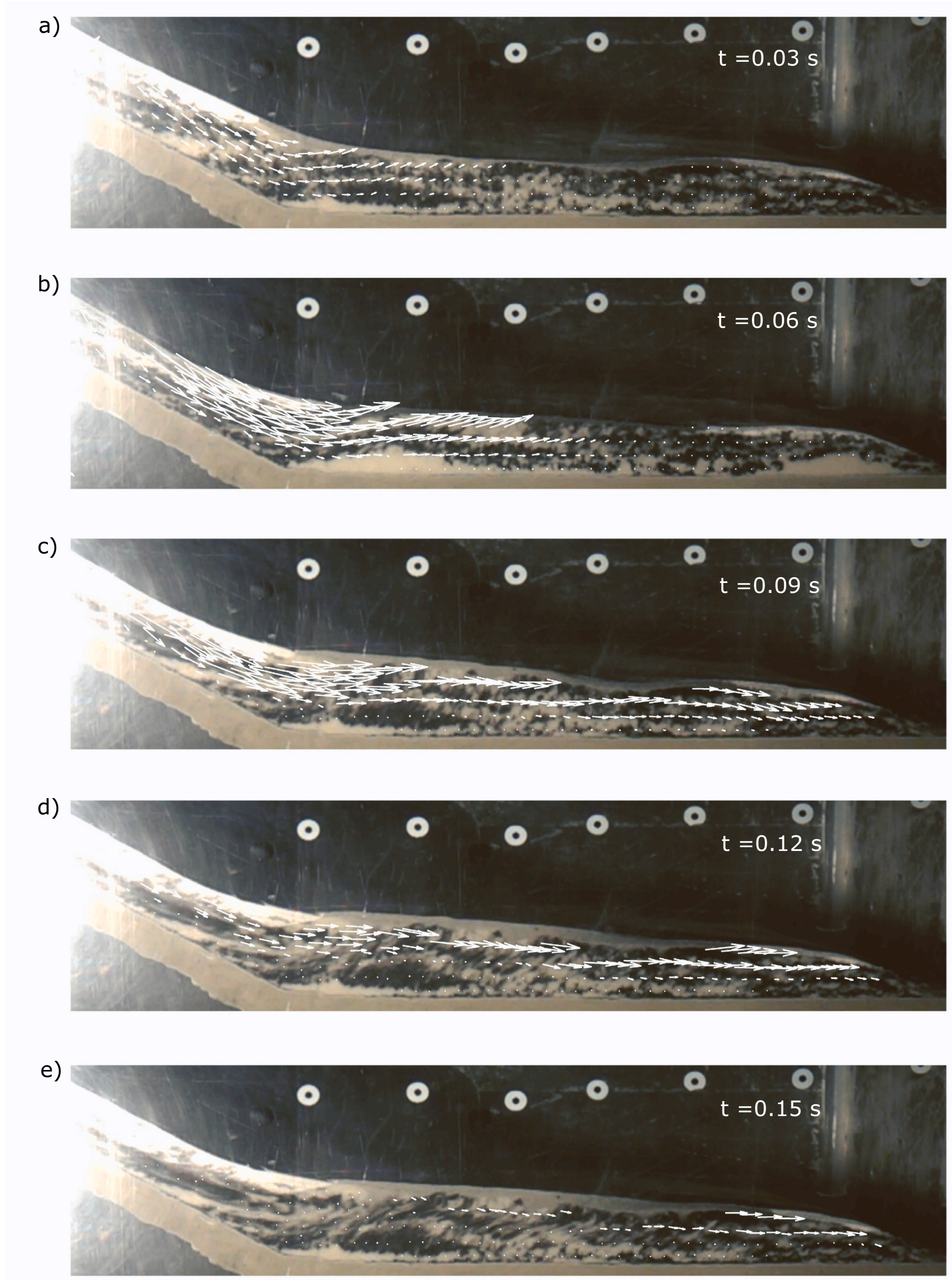

Fig. 6. Visual description of failure as captured using DIC analyses of high speed camera images taken every three one hundredths of a second during the triggering of instability and liquefaction in the valley floor sediments. The vectors represent the incremental displacement between time steps.

modelling investigating earthquake loading through the replacement of water as the pore fluid with a fluid of viscosity N-times higher. An excellent example of the consequences of this scale effect on liquefaction outcomes can be found in the work of Dewoolkar et al. (1999) who subjected level ground soil models saturated with water or a viscous pore fluid to identical earthquake motion. The magnitude of excess pore pressure during earthquake loading of the water saturated soil was observed to be reduced due to the high rate of pore pressure dissipation in relation to the rate of pore pressure generation. In contrast, the soil model saturated with the viscous pore fluid experienced higher excess pore water pressure and liquefaction under the identical earthquake loading.

Static liquefaction differs significantly from seismic liquefaction with regards to the time scale during which pore pressures are generated – within a single monotonic event rather than a gradual accumulation with an increasing number of cycles of earthquake loading. Considering the resulting implications for centrifuge modelling of static liquefaction events, Take et al. (2004) proposed that the time scale of generation of excess pore pressure is likely defined at the grain level as it is the instability of the soil skeleton that drives the monotonic increase in pore pressure towards static liquefaction. Further work exploring this hypothesis by Askarinejad et al. (2015) and Zhang and Askarinejad (2019) argues that the generation time of excess pore pressure due to the collapse of soil structure is $N^{0.5}$ times slower than its dissipation time at grain scale under Ng acceleration. In further work by Zhang and Askarinejad (2021) on submerged tilt-table tests exploring instability in

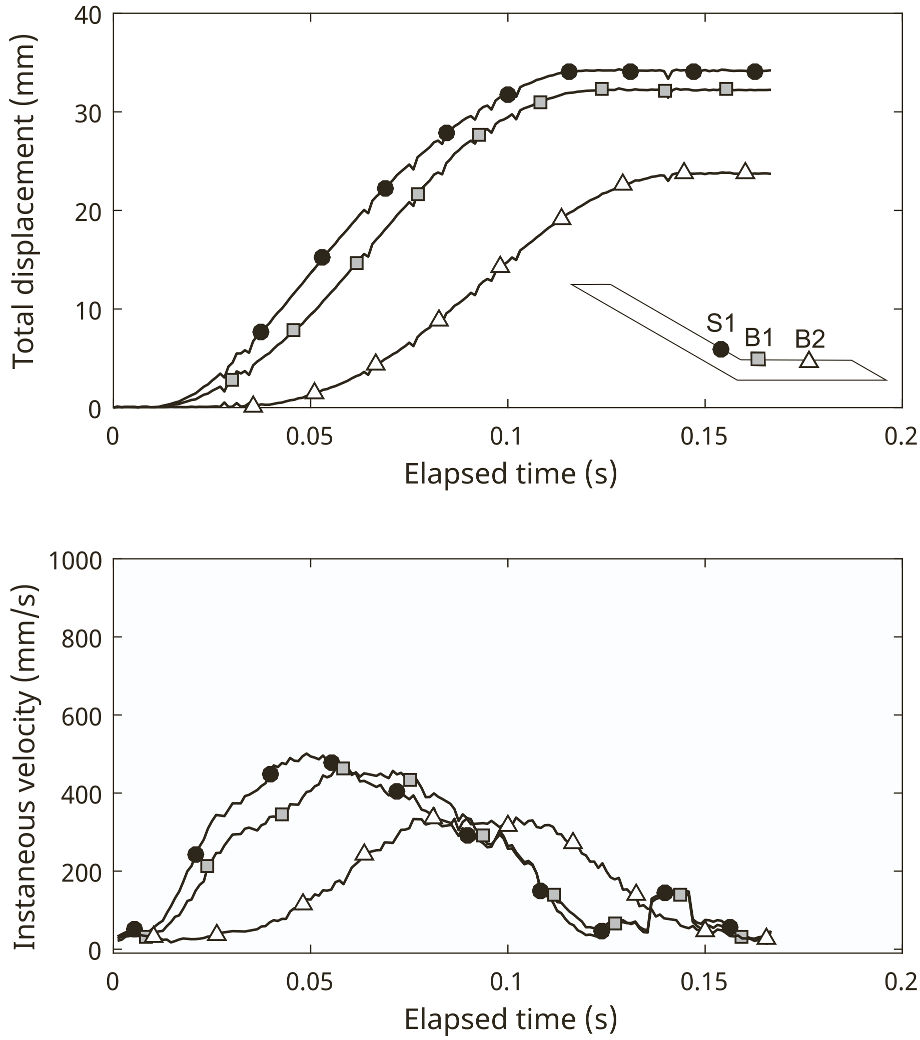

Fig. 7. Mobility of three DIC subsets defined on the slope (S1), and in the valley floor sediments proximal to the initial triggering toe failure (B1) and in the far-field (B2) in terms of displacement and velocity.

submarine landslides, experiments performed using fluid viscosity scaled as N results in a lower failure angle, i.e. a higher potential of static liquefaction. At the same g-level, they observed that N-fluid tests failed earlier than $N^{0.5}$ fluid tests. This study illustrates that a higher fluid viscosity reduces the capability of fluid dissipation, and thus boosts the process of changing soil drainage regime from drained to undrained conditions.

It is therefore impossible to have a centrifuge model in which the pore pressure response leading up to triggering (ie. transient seepage) and during pore pressure dissipation following liquefaction be identical to field scale. In the present work, the significance of this scale effect conflict was minimised through the use of a low acceleration level, $N = 20$, while retaining the practical and operational benefits of the use of water as the pore fluid (i.e. ability to spill water to the centrifuge chamber, no restrictions on the volume of fluid used to generate a steady state seepage regime, etc). Therefore, physical model tests exploring static liquefaction processes in which water is used as the pore fluid will dissipate pore fluid more rapidly and regain stability more quickly than the equivalent behaviour at field scale. In the context of this discussion on scale factors, it is important to reiterate that the focus of the present study is on the triggering of liquefaction in a single shear impulse from an overriding landslide, and not the post-liquefaction travel distance of the loose saturated base layer as this would be i) greatly impacted by the artificially fast rate of dissipation of pore water pressures through our decision to use water as the pore fluid, ii) subject to the additional complexities associated with bodies moving in a rotating frame of reference (see work on Coriolis effects as pertaining to landslide motion in geotechnical centrifuges by Bryant et al., 2015; Cabrera et al., 2020), and iii) runout distance significantly limited by the physical constraints of the test chamber. Thus, following other studies on static liquefaction in which water is used (e.g. Ng et al., 2022), the choice of water as a pore fluid should be viewed as creating a lower bound for modelling static liquefaction (e.g. if a viscous pore fluid were to have been used, any excess pore pressures generated because of a failure will take longer to dissipate, causing the slope to be even more unstable.)

2.2. Soil layer

A uniformly graded fine sand (#730 Silica Sand by Weldon Company), nominally identical to the F110 Ottawa sand used in past liquefaction studies (e.g. Take and Beddoe, 2014; Take et al., 2015) was selected for the soil layer. This soil was used by Beddoe and Take (2016) for a large landslide flume test, who found that the constant volume internal friction angles, ϕ', of the F110 Ottawa Sand and #730 Silica Sand were 31° and 30.5°, respectively. Additional details illustrating the comparison of the grain size distributions can be found in Beddoe and Take (2016). Prior to the construction of each model the sand was oven dried to remove any residual water. After drying, a prescribed mass of water was added to a known mass of soil to achieve a desired initial gravimetric water content of the soil to ensure the soil had a sufficient

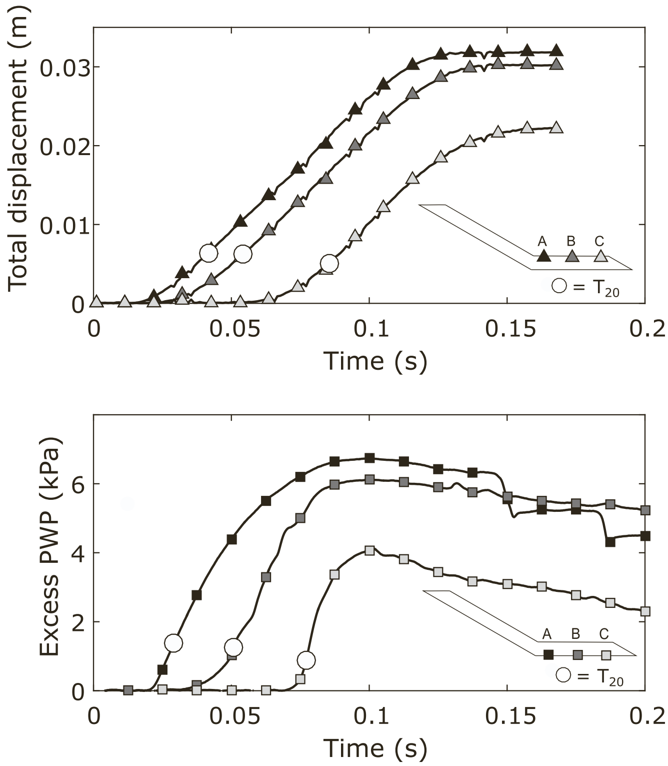

Fig. 8. Time of arrival of shear displacements and shear-induced excess pore water pressure as defined at the time at which the displacement or pore pressure exceeds 20% of its eventual maximum value at each x-coordinate corresponding to the location of a pore pressure transducer along the base of the model.

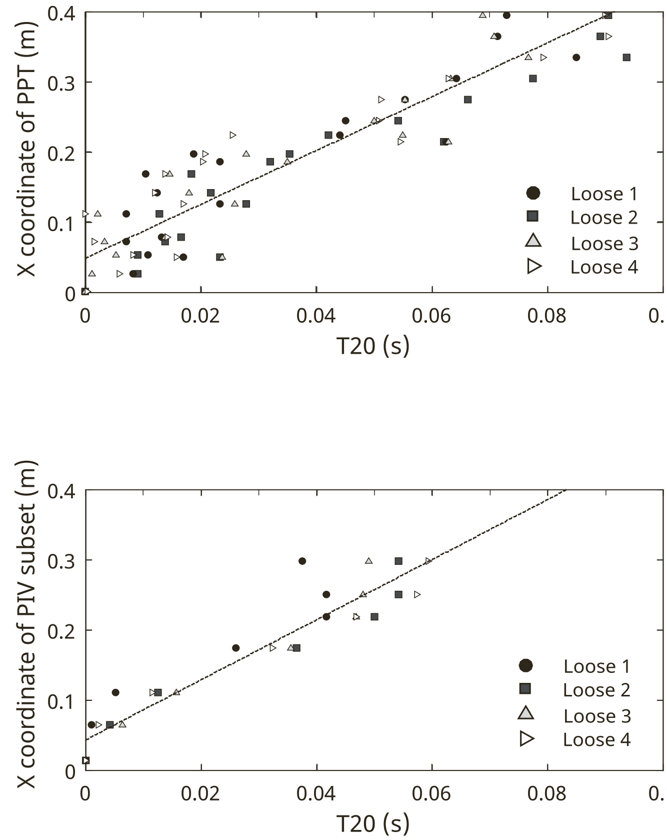

Fig. 9. Propagation velocity of liquefaction front calculated from time of arrival of shear displacements and shear-induced excess pore water pressure for four loose experiments. In all cases the best-fit velocity of 4.3 m/s (dotted regression line) is in excess of the 0.5 m/s velocity of the initial slope failure triggering instability.

matric suction to be placed in an extremely loose state. The moist soil was then rained through an ASTM No.5 sieve with 4 mm openings, controlling the drop height to achieve the target initial void ratio for each test. A vacuum nozzle was then used to remove any excess soil from the surface of the layer to ensure an even surface. Four loose soil models with void ratios in excess of 1.1, and two denser models with void ratios of 0.7 and 0.79 were created for testing (Table 1). The aim of including both contractive and dilative soil models in the experimental program was to explore the effect of initial state on the behaviour observed following landslide initiation.

2.3. Velocity measurement

A high-speed camera was included in the experiment for the measurement of the instantaneous velocity of all points in the landslide using the image processing technique of digital image correlation (DIC). This camera, Sony RX10M2, enabled the sequence of failure to be captured through the transparent side wall of the flume at 960 frames per second with a resolution of 1920 × 1080 pixels at g-levels up to 20 g. The camera was positioned to capture a large field of view (494 × 300 mm) observing both the portion of the slope that will be subjected to the initial triggering toe failure, and the entire length of the potentially liquefiable valley floor sediments to capture the velocity of instability in this layer (Fig. 3).

DIC software, such as the geoPIV code used in the present study (White et al., 2003), permits the tracking of incremental displacements of selected subsets between image frames, which when combined with the known time increment between frames, can produce measurements of instantaneous velocity. Given that the sand used in the present study was a uniform light colour, a small amount of sand was dyed black and selectively introduced at the transparent boundary to create a high-contrast image texture to improve DIC outcomes (e.g. Take, 2015).

2.4. Instrumentation to observe propagation of instability and liquefaction

While DIC can measure the relative velocity of the initial toe failure and the velocity of deformation in the valley floor sediments, measurements of the velocity of instability should ideally be based on the observation of shear-induced excess pore pressure. A unique strategy adopted in this paper is the installation of a dense linear network of 30 pore pressure sensors along the entire length of the model (i.e. at the location of each of the black squares denoted in the model geometry of Fig. 3) to permit the observation of a) the time at which shear-induced pore pressures were recorded, and b) the magnitude of the shear-induced pore pressure. The former, when combined with the known position of the sensor, permits a direct measurement of the velocity of instability as it propagates through the loose saturated layer. The latter, provides the ability to confirm whether the soil at each location liquefied.

As this dense sensor network approach would be cost prohibitive using traditional sensors used by the geotechnical physical modelling community (e.g. GE-Druck PDCR-81 pore pressure sensors), work was conducted to identify low-cost sensors capable of measuring pore pressures at a boundary. The sensor identified in this search, the Motorola MPX5050GP, were installed every 30 mm along the entirety of the slope and the base of the model (23 sensors in total). Prior to testing and after saturation of the sensors, a thin metal filter was placed over each sensor opening. This filter acted as a porous barrier allowing uninhibited flow of the water in the model to the sensor while simultaneously inhibiting any of the soil from clogging the sensor. To provide confidence in the readings of these sensors, seven GE-Druck PDCR-81 pore pressure sensors (i.e. the gold standard of pore pressure measurement in geotechnical centrifuge model tests) were placed adjacent to a low-cost sensor to ensure consistent results.

3. Static liquefaction experiments

3.1. Behaviour observed in loose tests

The behaviour observed in a typical loose slope experiment, Test Loose 1 (L1), is presented in this section. With an initial thickness of 52 mm, and a very loose metastable soil structure (e = 1.45), the soil layer experienced considerable settlement on delivery of the model to the testing acceleration of 20 g. The geometry of the soil layer at 20 g consists of a reduced thickness of 37.3 mm and is illustrated in Fig. 4a. Pumps providing the groundwater flux were then turned on and an initial groundwater table was introduced into the model. Fig. 4a presents the steady state total heads measured at the soil-bedrock contact. In other words, the pore water pressure data captured by each pore pressure transducer (PPT) is plotted as a total head at model scale above the x-location of each PPT. A few gaps exist in the dense network of PPTs spaced at 30 mm due to a higher loss rate of the low-cost sensors (i.e. using sensors not designed for inclusion in a damp location, resulting in electrical failure). As best illustrated at the lowest steady state groundwater flux (66 ml/min), the high hydraulic gradient along the sloping portion of the slope eliminates the possibility of the development of large flow thicknesses along the sloping portion of the soil layer. In contrast, the significantly reduced hydraulic gradient along the horizontal base permits the water to be retained in the soil before seeping out to the drain at the far right of the model. Stepwise increases in groundwater flux raise the groundwater level in the valley floor sediments until the phreatic surface intercepts the soil surface at the toe of the slope, immediately prior to failure. An additional increment of groundwater flux triggered a local failure of the toe. Image analysis of the failure will be presented later in the manuscript, but at this stage we will remain focused on the pore pressure response.

The typical response of the soil layer to shearing, beginning from a localized toe failure, and continuing as a propagating zone of liquefied soil is presented in terms of shear-induced pore pressure at two instants in time (0.07 s and 0.20s after triggering) in Fig. 4b and c, respectively. After 0.07 s after triggering, the region between x = 0 and 150 mm experiences significant shear-induced pore water pressures, while the valley floor sediments at x > 225 mm have yet to respond. A time increment of 0.13 s later, Fig. 4c, illustrates that this farther field region has begun to shear, while the earlier sheared region has already begun to experience pore pressure dissipation. In other words, Fig. 4a and b collectively indicate that a spatially propagating wave of shear-induce pore pressure is passing through the valley floor sediments. Later in the manuscript we will analyze the velocity of this wave; however, we shall first look at the typical magnitude of the shear induced pore pressures to confirm whether liquefaction has indeed occurred.

The pore water pressure ratio, r_u, defined as the ratio of pore water pressure to the total vertical stress at a point is a useful indicator of liquefaction. As r_u approaches 1, the effective confining stress approaches zero. Given these conditions, shear resisting forces drop significantly such that large accelerations can occur in a slope due to significant unbalanced forces. The pore pressure data observed in test L1 is presented in terms of r_u values in Fig. 5, with the pore pressure ratio at steady state seepage prior to failure at approximately 0.5. During the failure event, large r_u values (0.8 to 1) signify that the base did liquefy, and that the liquefaction extended along the majority of the base.

A visual description of the failure process of a typical loose test is presented in Fig. 6 for Test L1 in increments of three one hundredths of a second. During the first 0.03 s, the incremental displacement vectors illustrate a localized rotational failure at the toe of the slope (Fig. 6a). This then increases in speed as illustrated by the size of the vectors in Fig. 6b, and the zone of failure rapidly increases until the entire horizontal loose saturated soil layer is shearing (Fig. 6d). With pore pressure dissipation, the zone of soil where failure initially began ceases movement prior to the shearing front coming to rest in the far field of the horizontal soil layer (Fig. 6e). Given that the motion of the landslide in the loose saturated basal layer is predominantly horizontal, the increase in pore water pressures observed along the base of this layer occur at near constant soil thickness, are not simply the result of additional height of material accumulating over the sensors, but rather are indicative of shear induced excess pore water pressures.

The typical displacements of individual locations in the landslide affected soils on the slope and horizontal valley floor sediment layer are illustrated in Fig. 7. The displacement of the slope region whose location is illustrated in the inset figure of Fig. 7a as S1, is part of the initial toe failure and begins moving first and displaces at total of 33 mm in model scale. A region of the base, B1, located within this initial failure experiences the same behaviour. However, a region 100 mm away, B2, that is out of the reach of the 33 mm distance of the initial toe failure experiences shearing, indicating that propagation velocity of instability is higher than the velocity of the toe failure. Velocities calculated using DIC of these three subsets are shown in Fig. 7b, confirming that the velocity of the toe failure was on the order of 0.5 m/s. In the next section we will compare this to the propagation velocity of shear induced pore pressure.

To compare the velocity of propagation of the liquefaction wave in the valley floor sediments, DIC subsets were defined at the soil surface every 30 mm along the base of the flume to be coincident with the location of pore pressure sensors. The arrival time of the shear wave could then be defined as the time of arrival of either flow displacement from DIC or time of rapid generation of shear-induced pore water pressure. In this paper, the authors used a strategy where the arrival time in each case was defined as the time at which the value of displacement or pore pressure exceeds 20% of its eventual maximum value. The selection of 20% (as compared to 10% or 30% for example) was done based on the observation that by this time in each test, a constant velocity of the flow and rate of excess pore water pressure generation had been reached. This selection process is illustrated in Fig. 8 for three locations along the length of the horizontal layer, with the t_{20} value, illustrated as a large open circle.

The arrival time of either the flow displacement or the onset of rapid generation of shear-induced pore pressure can then be plotted against the x-coordinate of the sensing location to define the propagation velocity of the shear wave. Data from all four loose tests, with slightly varying initial thicknesses and void ratios presented in Fig. 9, indicate a velocity of propagation through all of the models of 4.2–4.3 m/s. Comparing this rate to the velocity of the initial failures (0.5 m/s), it can be seen that, for the conditions tested, the instability front travels at a velocity an order of magnitude larger than the initial triggering event. This data confirms the hypothesis that, in certain circumstances, the velocity of instability can exceed the velocity of any one location in the landslide. In other words, the velocity of growth and entrainment of the failure volume can exceed the maximum velocity of any one location within the landslide.

3.2. Behaviour observed in dense tests

Pre-failure instability is a phenomenon of loose, saturated granular materials. As an illustration of the importance of the contractile nature of the soils used in the present study for the mechanism observed, two additional nominally identical slopes were created with sole exception being that the soil was placed having sufficient density to be dilative. Groundwater flow was continuously increased to try and induce a failure event; however, no such failure had yet been reached when the pumps reached their maximum flowrate. The final flow rates for the two dense tests were greater than that of all the loose tests (Table 1).

4. Discussion

The results of the experiments showed that the initial velocity of the landslide was an order of magnitude slower than the subsequent velocity of instability. In other words, these experiments illustrate that the velocity of instability can significantly exceed the speed of the initial failure under the conditions tested. Due to the limitations of the physical model, the liquefaction of the base stopped at the end of the model once there was no more soil to liquefy. It is therefore possible that the zone of liquefaction at the base of the landslide could have extended farther if the dimensions of the model were longer.

These observations have significant implications for the prediction of the maximum distal reach of landslides in which static liquefaction may play a role. Most dynamic models of landslide runout are calibrated using rheological parameters such that the release of a soil volume on the topography matches the trimline of the observed failure, with entrainment only occurring under the sliding mass. However, the experimental observations presented in this paper indicate that, in certain circumstances, the landslide front can expand at a velocity an order of magnitude higher than the initial failure – essentially liquefying the base soil in front, rather than simply underneath the initial localized toe failure.

Further, these observations illustrate that a localized failure onto a liquefiable deposit may propagate liquefaction much farther than simply the runout of the localized failure. Such a mechanism may have played a significant role in the failure of 600 m of the Prat quay wall in Barcelona Harbour on January 1st 2007. In this case study, the quay wall in question consisted of caissons founded a rubble mound backfilled by hydraulic fill. During the failure event the hydraulic fill liquefied and provided the necessary thrust to displace the caissons up to 90 m into the harbour. As reported by Gens (2019), "After investigation, it is strongly suspected that failure of the South embankment started the backfill liquefaction that subsequently spread to other locations." Further analysis of this event by Tarragó (2021) explored two possible triggering mechanisms that can explain the failure: spontaneous liquefaction following continuing fill placement, or liquefaction occurring locally at the tip of the South embankment (under construction at the time of failure) and propagating through the hydraulic fill mass towards the caisson wall. Our observations of a localized failure into a liquefiable deposit may propagate liquefaction much farther than simply the runout of a single localized failure, thereby providing additional support to the hypothesis that the failure initiated by an internal embankment at Prat quay could have triggered a more widespread failure.

5. Conclusions

Four landslide experiments with a loose contractile soil and two experiments with a dense dilative soil were performed in a geotechnical centrifuge. The landslides were triggered under rising groundwater conditions to initiate landslide motion to override a layer of soil at the base of a landslide. In the four loose saturated soil layer experiments, the overriding landslide caused the saturated soil layer to experience instability leading to static liquefaction, whereas the experiments performed on dense soil did not. A novel experimental technique using a dense linear sensor network of pore water pressure sensors was used to capture, for the first time, the propagation velocity of instability through the loose saturated deposit impacted by the initial landslide. The velocity of this initial failure (0.5 m/s) was approximately an order of magnitude slower than the subsequent 4.2 m/s propagation velocity of the liquefaction front. This surprising observation leads to new scientific questions being raised relating to the exact nature of the mechanism of instability: What are the physical properties of the soil, characteristics of the liquefaction event, and geometric controls that define the rate of propagation of liquefaction in loose saturated sediments? Does this velocity of instability represent an upper limit to the rate of liquefaction of such a layer? In other words, does this velocity define a threshold that separates behaviour based on the initial velocity of the triggering failure? For example, in scenarios in which the velocity of the initial triggering event is lower than the propagation velocity of instability (e.g. the failure of a low height internal dyke into loose saturated tailings or hydraulically placed fill) the centrifuge test results indicate the propagation of instability may spread faster and farther than might be expected from a conventional runout analysis. However, in scenarios in which the velocity of the initial failure significantly exceeds the velocity of instability, the initial sliding mass may either simply override these sediments or result in a more complex spatially and temporally varying interaction between the initial sliding mass and the liquefiable sediments. The observations of instability in the physical model tests presented in this study illustrate the complexity of the physics behind predictions of landslide runout in which instability and static liquefaction are triggered, and highlight remaining questions that must be resolved through future work.

CRediT authorship contribution statement

Liam J. Steers: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Ryley A. Beddoe:** Writing – review & editing, Supervision, Project administration, Conceptualization. **W. Andy Take:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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